

Project Development Phase

Coding & Solution

Date	10/07/2026
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Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution:

This section evaluates the implementation of the **Credit Card Approval Prediction** project. The solution is developed using Python and machine learning libraries to predict whether a credit card application should be approved based on applicant details. The project follows clean coding practices, modular programming, proper documentation, and error handling to ensure maintainability and scalability.

Instructions:

1. The project code is modular, well-structured, and follows Python coding standards.
2. The solution successfully predicts credit card approval using trained machine learning models.
3. All required source files, trained model files, datasets, and dependency files are included.
4. The project can be maintained using a Git repository for version control.

Solution Summary:

Field	Details
Repository Link / URL	
Programming Language(s)	Python 3.x
Framework(s) Used	Flask (Web Application), Scikit-learn, Pandas, NumPy, Joblib
Key Features Implemented	Data preprocessing, Feature engineering, Model training, Credit card approval prediction, Model evaluation, User-friendly web interface
Pending / Incomplete Features	Real-time API integration, User authentication, Cloud deployment (optional enhancements)
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , run <code>app.py</code> , and access the application through the local web browser.

Code Quality Checklist:

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (<i>if using GitHub</i>)

Additional Notes / Comments:

The project follows a modular architecture by separating data preprocessing, model training, prediction, and evaluation into independent modules. Standard Python naming conventions (PEP 8) are followed to improve readability and maintainability. Scikit-learn pipelines are used for efficient preprocessing and prediction. The trained machine learning model is saved using **Joblib**, allowing quick loading for future predictions without retraining. The project is designed to be easily extendable by incorporating additional machine learning algorithms, larger datasets, or deployment on cloud platforms such as AWS, Azure, or Heroku.